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/*****/
// CHÚC NĂNG: RÁNH VẬT CẢN
/*****/

#include <Wire.h> //Includes I2C Lib
#include <NewPing.h>
#include <Servo.h>

#define ECHO_PIN 12 //chân ECHO của siêu âm - nối 12 Shield V5.0
#define TRIG_PIN 13 // chân ECHO của siêu âm - nối 13 Shield V5.0
#define MAX_DISTANCE 200
#define MAX_SPEED 235 // sets speed of DC motors
#define MAX_SPEED_OFFSET 20

Servo myservo;

NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE);
boolean goesForward=false;
int distance = 100;
int speedSet = 0;

int      dugme=1;

// Nối chân L298N với shield V5.0
const int enAPin = 6; // (V5) chân 6 - ENA (L298N)
const int in1Pin = 7; // (V5) chân 7 - IN1 (L298N)
const int in2Pin = 5; // (V5) chân 5 - IN2 (L298N)
const int in3Pin = 4; // (V5) chân 4 - IN3 (L298N)
const int in4Pin = 2; // (V5) chân 2 - IN4 (L298N)
const int enBPin = 3; // (V5) chân 3 - ENB (L298N)

int command; //bien lưu trạng thái Bluetooth
int speedCar = 220; // tốc độ 50 - 255.
int speed_Coeff = 4; // hệ số suy giảm <=> giảm máy lan tốc độ

void setup() {

    pinMode(enAPin, OUTPUT);
    pinMode(in1Pin, OUTPUT);
    pinMode(in2Pin, OUTPUT);
    pinMode(in3Pin, OUTPUT);
    pinMode(in4Pin, OUTPUT);
    pinMode(enBPin, OUTPUT);

    Serial.begin(9600); //Init Serial port rate set to 9600

    myservo.attach(11); //servo nối chân số 11 Shield V5.0
    myservo.write(95); //Góc ban đầu Servo, nếu nó lệch (không
    thẳng) bạn chỉnh giá trị khác tăng hoặc giảm
    delay(2000);

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        distance = readPing();
        delay(100);
        distance = readPing();
        delay(100);
        distance = readPing();
        delay(100);
        distance = readPing();
        delay(100);
    }

    void loop() {

        long duration, distance;
        int distanceR = 0;
        int distanceL = 0;
        while (dugme==1)
        {
            distance = readPing();
            delay(40);
            if (distance<=25)
            {
                stopRobot();
                delay(100);
                goBack();
                delay(200);
                stopRobot();
                delay(200);
                distanceR = lookRight();
                delay(200);
                distanceL = lookLeft();
                delay(200);
                if (distanceR>=distanceL)
                {
                    goRight();
                    delay(300);
                    stopRobot();
                    delay(100);
                }else
                {
                    goLeft();
                    delay(300);
                    stopRobot();
                    delay(100);
                }
            }else
            {
                goAhead();
            }
        }
    }
}

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void goAhead() {

    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goBack() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);
}

void goRight() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);
}

void goLeft() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);
}

void goAheadRight() { //vua tien - vua quay phai
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon
}

void goAheadLeft() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien

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        digitalWrite(in3Pin,LOW);
        digitalWrite(in4Pin,HIGH);           // RIGHT tien
        analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham
hon
        analogWrite(enBPin, speedCar);

    }

void goBackRight(){           //vua lui - vua quay phai
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);           // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon

}

void goBackLeft() {
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);           // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham hon
    analogWrite(enBPin, speedCar);

}

void stopRobot(){
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,LOW);
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

int lookRight()
{
    myservo.write(40);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(95);           //cái này chỉnh cho thẳng
    return distance;
}

int lookLeft()
{
    myservo.write(150);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(95);           //cái này chỉnh cho thẳng
}

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        return distance;
        delay(100);
    }
int readPing() {
    delay(70);
    int cm = sonar.ping_cm();
    if(cm==0)
    {
        cm = 250;
    }
    return cm;
}
```